

The Hybrid Automation Playbook

How Regulated Industries Eliminate Manual Data Entry with Zero Compliance Risk

The Real Bottleneck Isn't the AI. It's Where the Data Has to Go.

Every regulated organization has the same problem: critical information — contracts, claims, patient records, closing documents — sits locked inside disconnected systems, and getting it out means manual re-keying, manual cross-checking, and manual routing. The cost isn't just time. It's the errors that slip through when a human has to copy the same number between four applications, and the exposure that piles up every time sensitive data changes hands to get processed.

Most automation tools "solve" this by asking you to trust a single black-box cloud. For a bank, a hospital, or a law firm, that's not a real answer — it's a non-starter dressed up as a feature.

Flow takes a different position: automation should adapt to your compliance posture, not the other way around.

THE CORE IDEA

Deployment Should Match Your Compliance Posture

Flow runs the same Action Graph engine — the same document understanding, the same workflow logic, the same human-in-the-loop approvals — regardless of where it's deployed. What changes is *where the processing happens*, and that choice belongs to your IT and compliance team, not to us. Flow ships in two deployment models:

MODEL 1

Secure Cloud

Flow's dedicated, secure cloud environment. Sensitive fields — PII, account numbers, policy data — are automatically detected and masked before any processing takes place.

MODEL 2

Flow-Edge / Flow-OnPrem

For organizations whose internal policy or regulation requires it: data stays on the workstation, or on a dedicated server inside the organization's own internal network. Nothing crosses the network boundary.

Same engine, two ways to run it. The rest of this guide shows what that looks like in practice, across three regulated sectors.

— THREE SECTORS, TWO ARCHITECTURES

Finance & Insurance — Automated Claims Routing & Expense Auditing

SECURE CLOUD

Claims and expense line items arrive faster than teams can route and audit them — every item needs to be checked against policy limits, matched to the right adjuster or cost center, and flagged the moment something breaches a threshold. Every manual touch is another chance for a transcription error to become a payout error.

Flow ingests claims and expense reports directly from your existing systems. Account numbers, policyholder identifiers, and other sensitive fields are automatically masked *before* the document ever reaches Flow's cloud environment — the AI processes what it needs to route and audit the claim without ever seeing the raw sensitive data. Flow then routes each claim to the correct queue and flags any expense line item that breaches policy for human review. You get cloud-grade speed and scale, with masking doing the compliance work under the hood.

Healthcare — Local Ingestion of Patient Charts & Medical Records

FLOW-EDGE / FLOW-ONPREM

Under HIPAA, PHI can't leave the hospital network — full stop. But charts and records still need to be read, summarized, and pushed into EHR workflows every single day, and "we can't automate this because of compliance" has been the default answer for too long.

Flow-Edge changes that answer. The entire pipeline — OCR, extraction, summarization, routing into the EHR — runs on a workstation or dedicated server inside the hospital's own network. Patient charts and medical records are read and acted on without ever traversing an external network boundary. Same document understanding, same human approval checkpoints as the cloud model — the only difference is that the data never leaves the building to get there.

Legal & Compliance — Cross-Checking Transactional Files & Automated Closing Milestones

FLOW-EDGE / FLOW-ONPREM

Real-estate and M&A closings depend on cross-referencing tax records, title documents, and closing ledgers across multiple parties — a single missed cross-reference can mean a multi-million-dollar transaction stalls, or worse, closes on bad numbers.

This is the exact workflow behind the live demo on our site: a Tax Clearance Certificate arrives for **Project Ironwood**, and Flow — running on a dedicated server inside the firm's own network — automatically detects it, matches it to the active transaction by parcel ID, and cross-checks it against the pending escrow estimate. After a one-click human approval, Flow updates the Transfer Deed with the new tax receipt, recalculates the Closing Statement, and produces the final **Cash to Close** figure (\$1,240,500) — then drafts the client dispatch email for a second approval before it goes out. Three human approvals, zero manual data entry, and the firm's transaction data never leaves its own infrastructure. If you've already run the demo, this is that workflow, running on Flow-OnPrem, for real.

CHOOSING YOUR ARCHITECTURE

Two Models, One Engine

There is no universally "right" deployment model — only the one that matches your regulatory posture and existing infrastructure. Both run the same Flow engine; the two options below are equally supported, and many organizations run both at once, matched to each workflow's own sensitivity.

	Secure Cloud	Flow-Edge / Flow-OnPrem
DATA LOCATION	Masked data processed in Flow's dedicated secure cloud	On the workstation or a dedicated server inside your own internal network
BEST FOR	High-volume workflows where speed and scale matter most	Workflows where internal policy or regulation requires data to stay in-network
COMPLIANCE POSTURE	Automatic masking before any processing	No data ever crosses the network boundary

The right architecture for your organization — or mix of architectures across different workflows — depends on specifics: your existing infrastructure, your regulator, your risk tolerance. That's exactly what a discovery call is for. We'll walk through your environment, map it to the model (or models) that fit, and show you what it looks like running on your own data.

Ready to see which model fits your organization?
Book a discovery call.